POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Embedded System

Semester: Spring

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

)	What is embedded system? Explain the components of embedded hardware.	7
)	Explain with an example how to optimize custom single purpose processors.	8
)	Draw the combinational logic design for three inputs a, b & c and two outputs y & z. The output 'y' is such that y is 1 if a is 1 or b & c is 1 and z is 1 if b or c is 1 but not both.	7
)	How does a programmer view a microprocessor based embedded system? What are his/her concern?	8
)	Compose $2^{(k+1)}$ X 2n memory using 2^k X n memory modules.	7
)	What is interrupt? Explain the steps used in data transfer using vector interrupt along with its flowchart.	8
)	Explain any two arbitration techniques that implemented to communicate with peripheral devices from the microprocessor.	7
)	List out the difference between process and thread? Explain various state of process.	8
)	Define Real Time Operating System. Differentiate between clocking communication and task synchronization.	7
)	Explain briefly the architecture of 8051 microcontroller with the aid of block diagram.	8
a)	Give an introduction to VHDL and explain the basic structure of a VHDL file with examples.	7
b)	Write a VHDL program for which output will be 1 when the sequence 101 is detected.	8

Write short notes on: (Any two) 2×5

a) Task, task states and task scheduling
b) Cross compiler and cross assembler
c) VHDL program to simulate binary adder

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